failure2 — Native-IMU reconstruction: method and interpretation

Separate experiment: existing mocap identification and outputs are unchanged.

Identification uses native gyro / specific force with fitted thrust, gyro and accel first-order lags.

Orientation: gyro prediction + acceleration update recur; magnetic factors affect output only.

Only initial orientation / position and initial velocity use mocap. Later mocap is read-only evaluation.

Comparison uses original mocap timestamps inside IMU support; reconstructed attitudes are Slerped.

Native timestamps and firmware first-or-changed magnetic freshness proxy are retained.

Static sensor covariance = trace(unbiased static covariance)/3 times identity; no bias subtraction.

Offline local polynomial derivatives and their overlap covariance are included in the lag fit.

Uncertainty: Bingham-derived right-angle covariance and deterministic local Gaussian propagation.

Shared acceleration/attitude information uses declared conservative covariance bounds.

These algebraic bounds do not guarantee coverage of systematic acceleration or magnetic bias.

Magnetic output can be worse on these data; no mocap-based gating, tuning or recursive feedback.

Model reconstruction adds the native-IMU residual wrench time series to the physical dynamics.

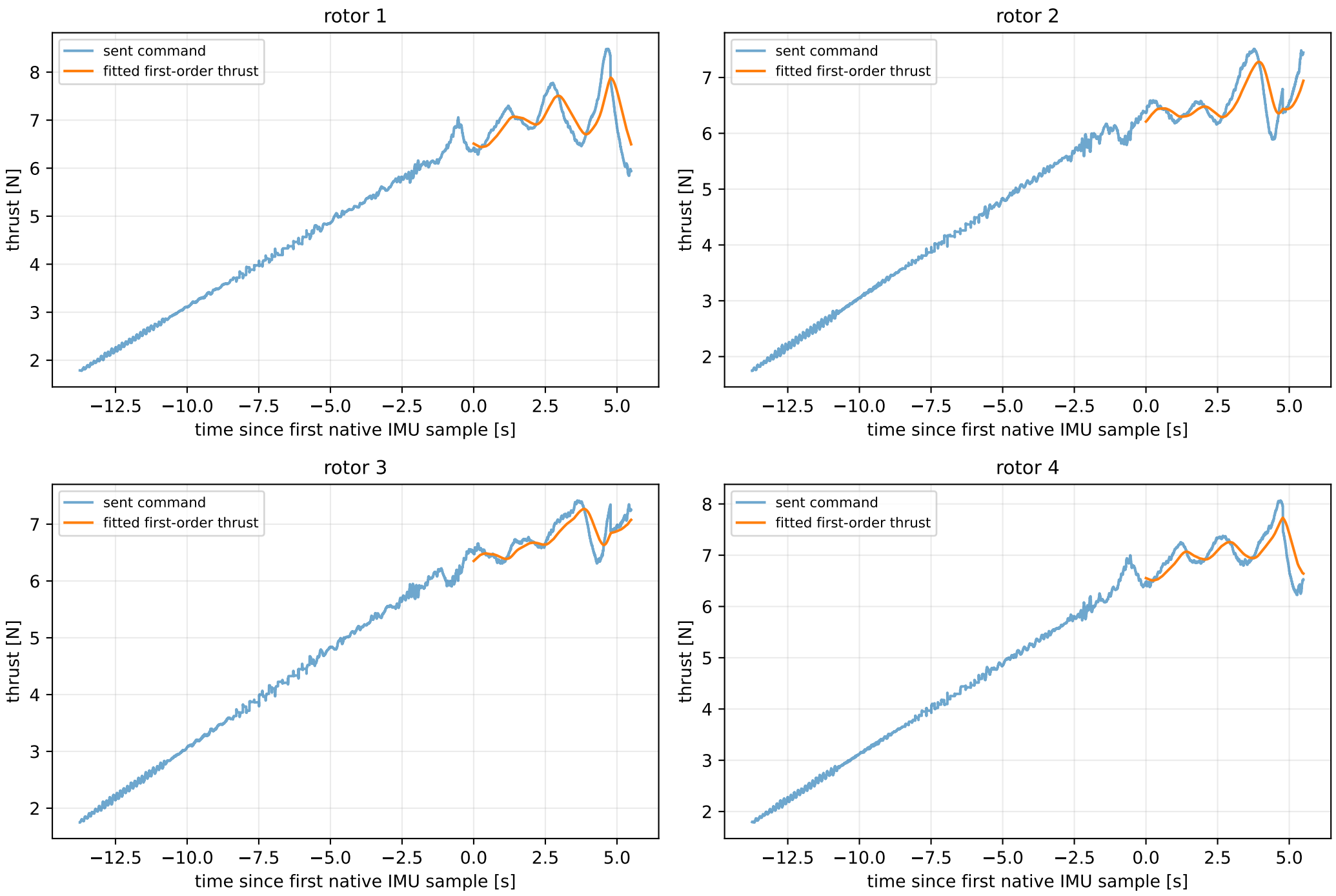
No Monte Carlo, artificial display variance, eigenvalue clipping or uncertainty-based sample selection.

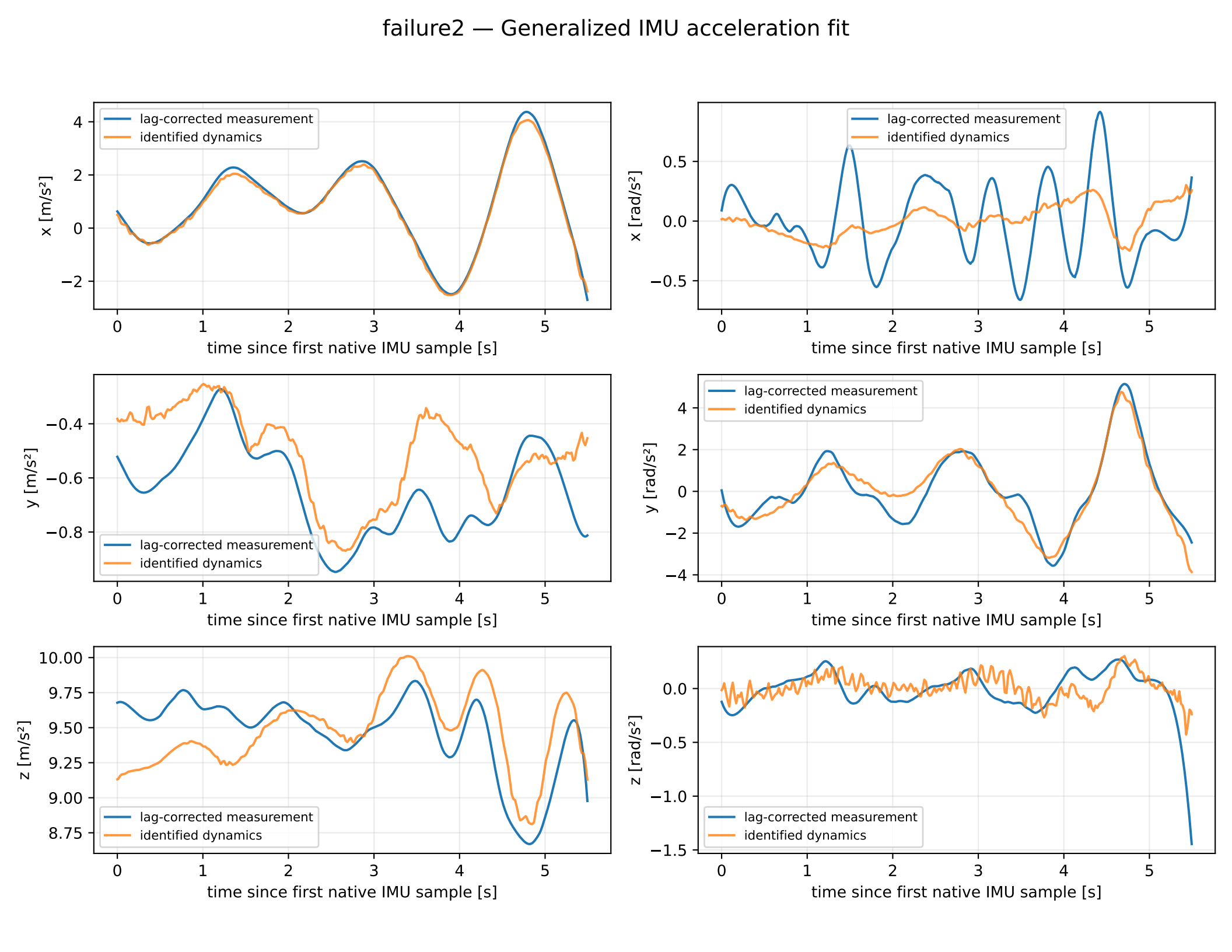
Mass / inertia / thrust scale use a common-scale gauge; chip lever arm is fixed, not calibrated.

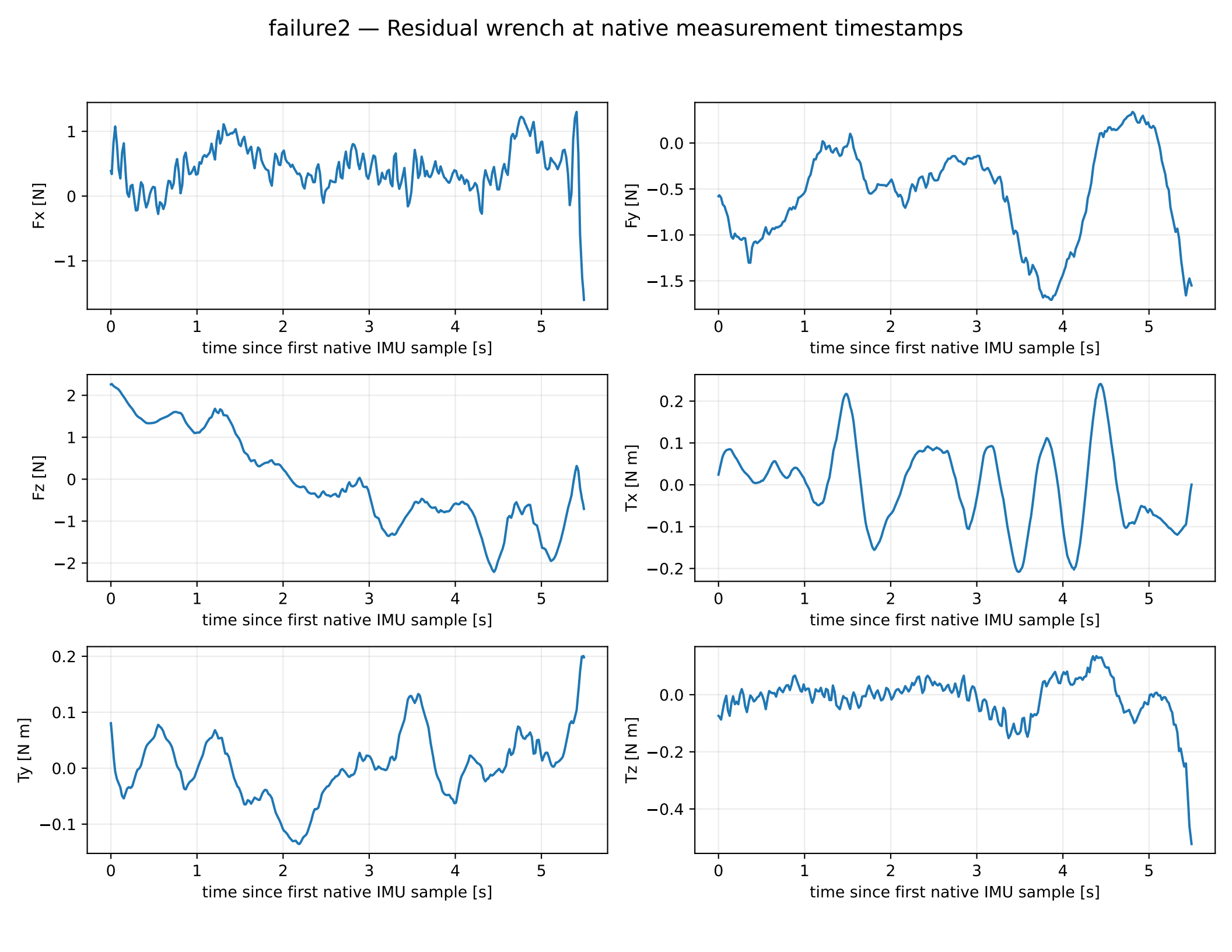
All 17 actual SI estimates and the complete joint covariance are appended at the end.

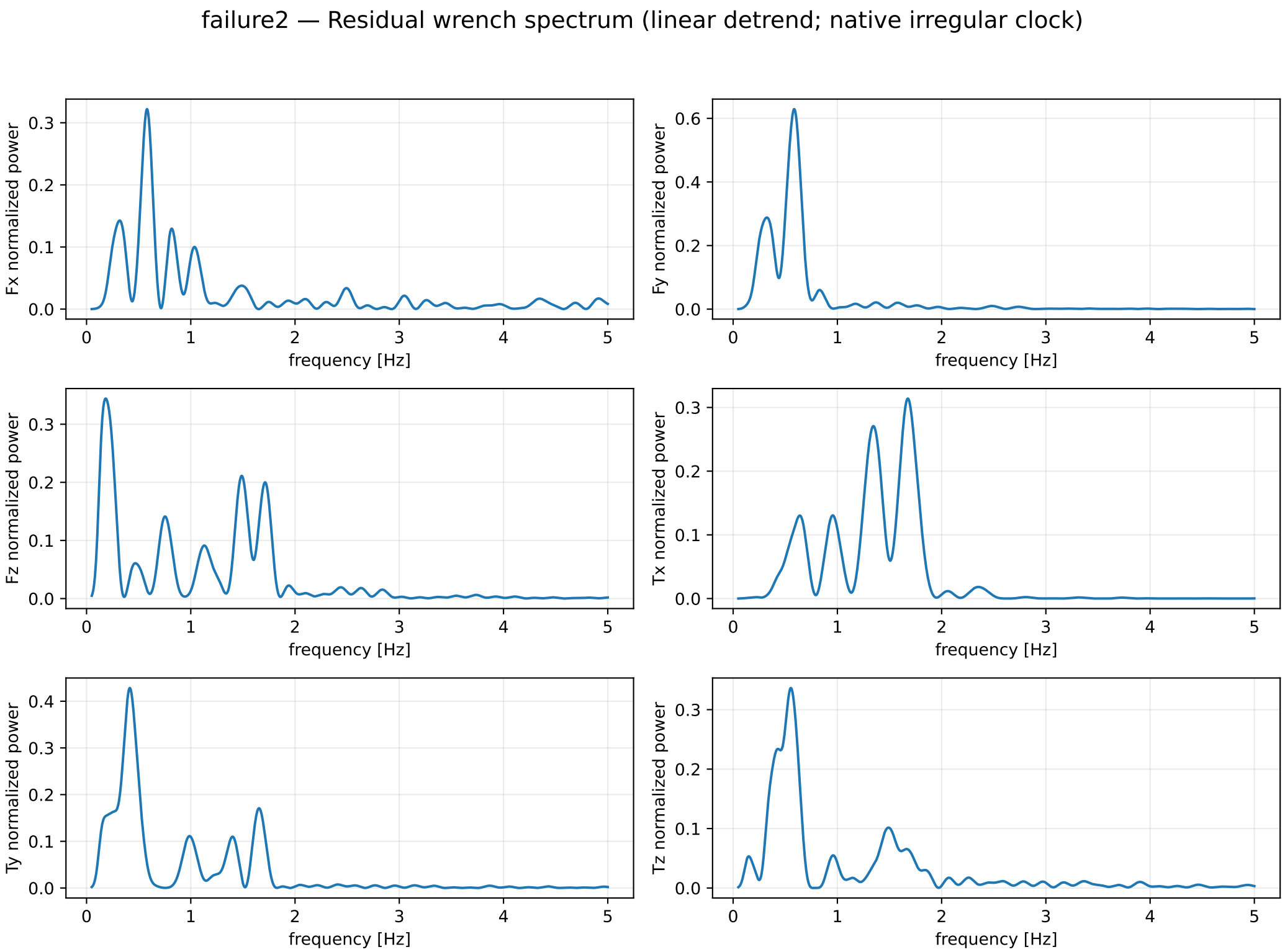
Native IMU support: 25.504054 to 30.998216 bag seconds; N = 1100.

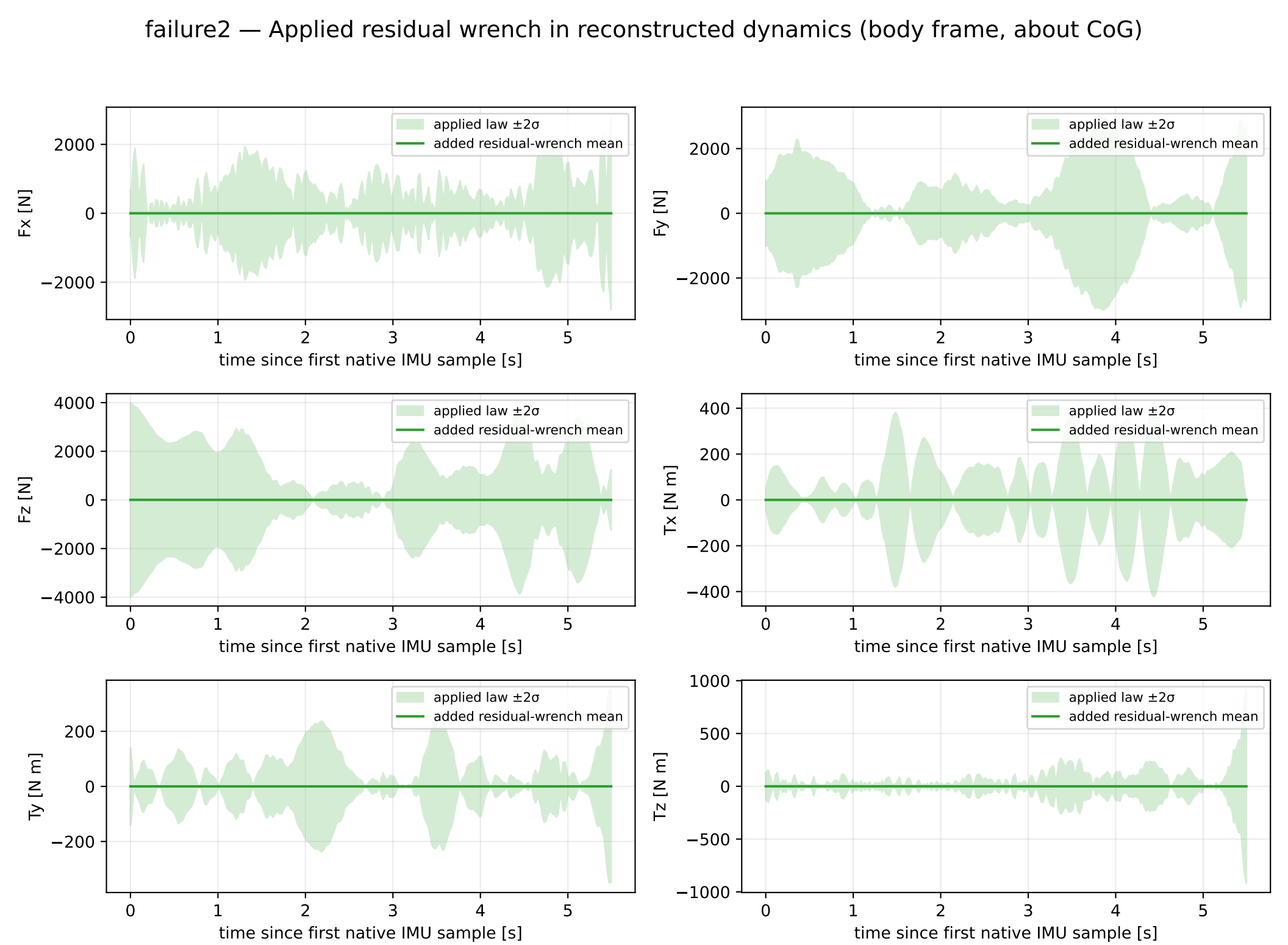
Individual PDF appendices and full-precision covariance CSV contain the same distribution.

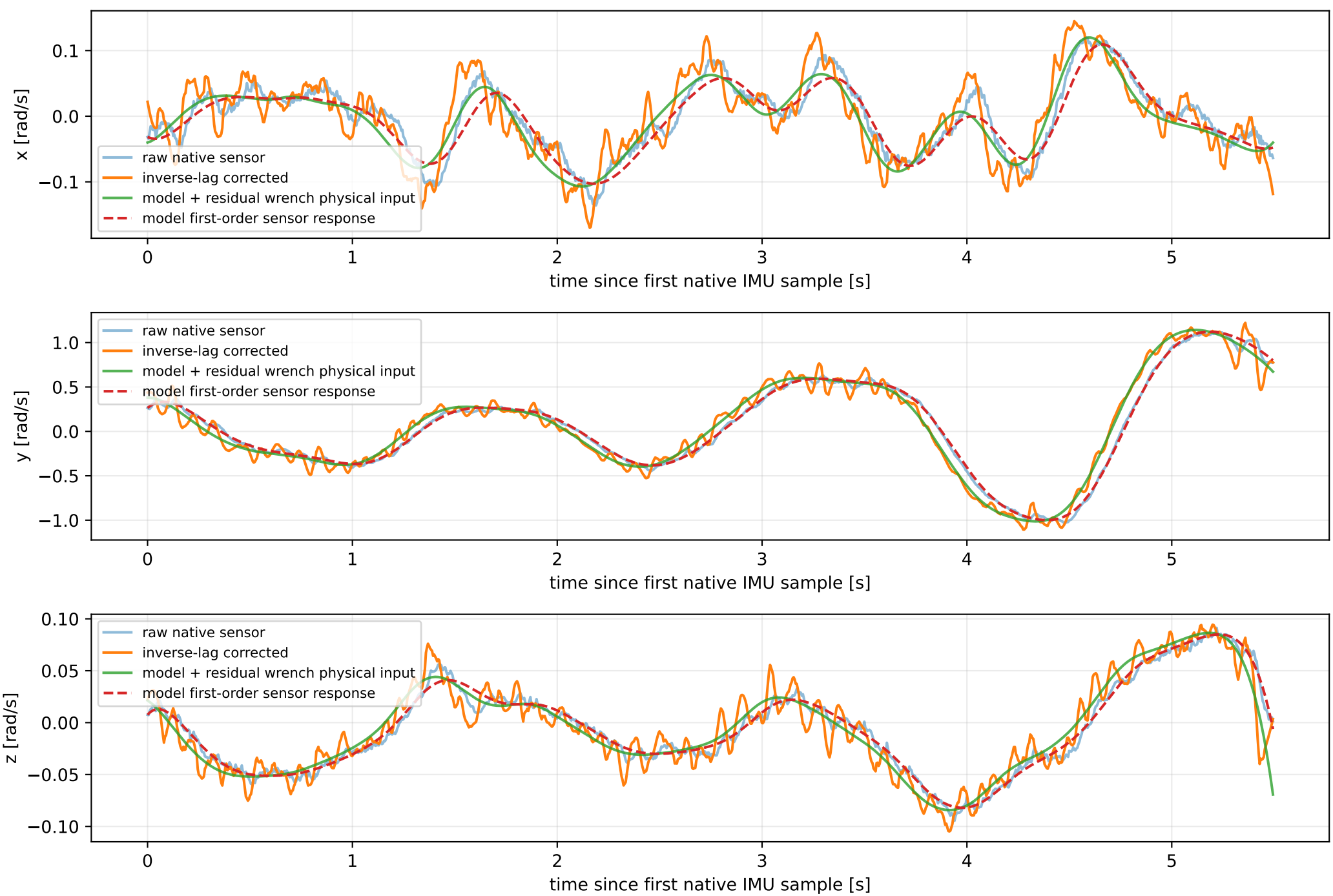

failure2 — Rotor commands and fitted thrust dynamics
rotor 1
rotor 2
rotor 3
rotor 4
sent command
fitted first-order thrust
thrust [N]
time since first native IMU sample [s]
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0

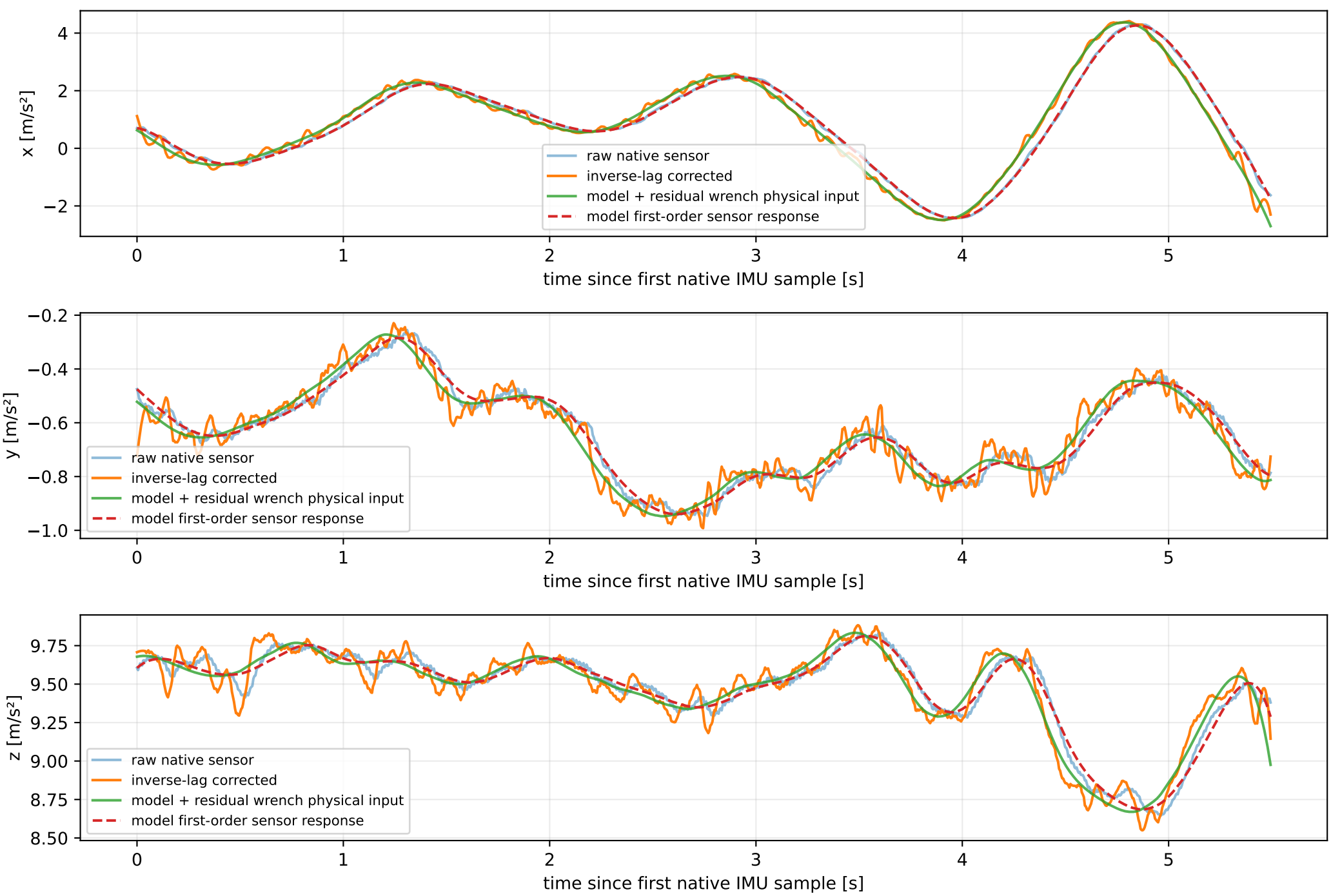

failure2 — Generalized IMU acceleration fit
lag-corrected measurement
identified dynamics
x [m/s²]
x [rad/s²]
y [m/s²]
y [rad/s²]
z [m/s²]
z [rad/s²]
time since first native IMU sample [s]

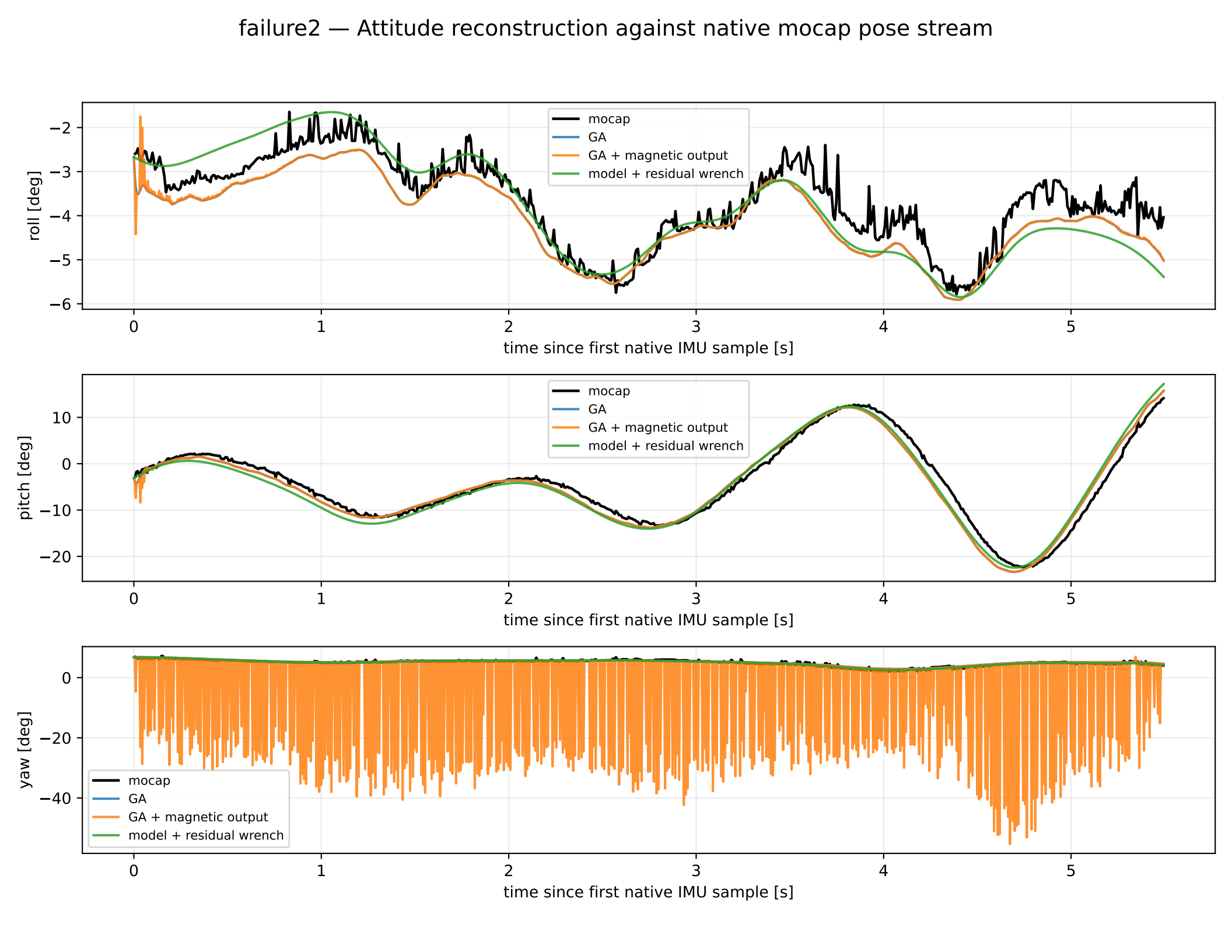

failure2 — Residual wrench at native measurement timestamps
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time since first native IMU sample [s]

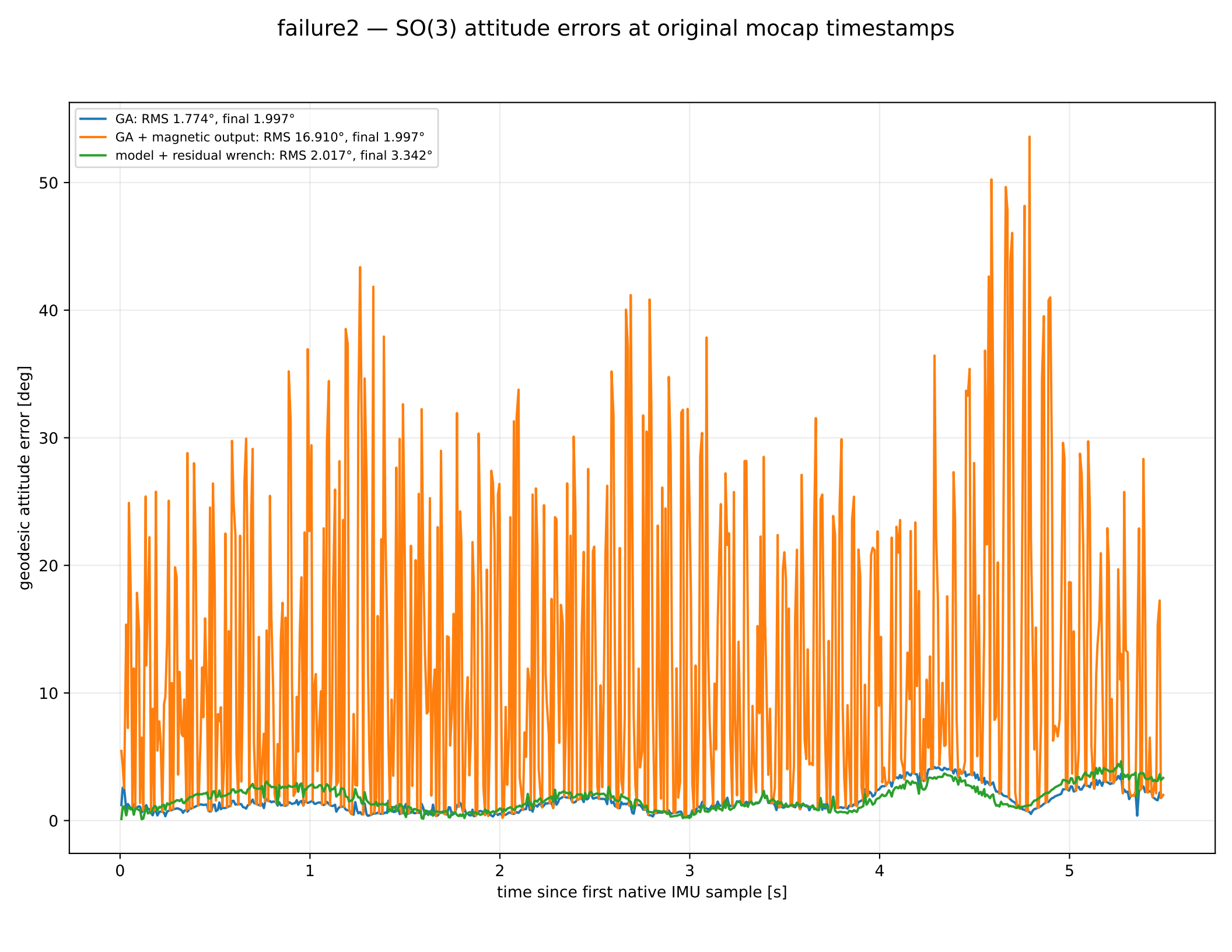

failure2 — Residual wrench spectrum (linear detrend; native irregular clock)
Fx normalized power
Fy normalized power
Fz normalized power
Tx normalized power
Ty normalized power
Tz normalized power
frequency [Hz]

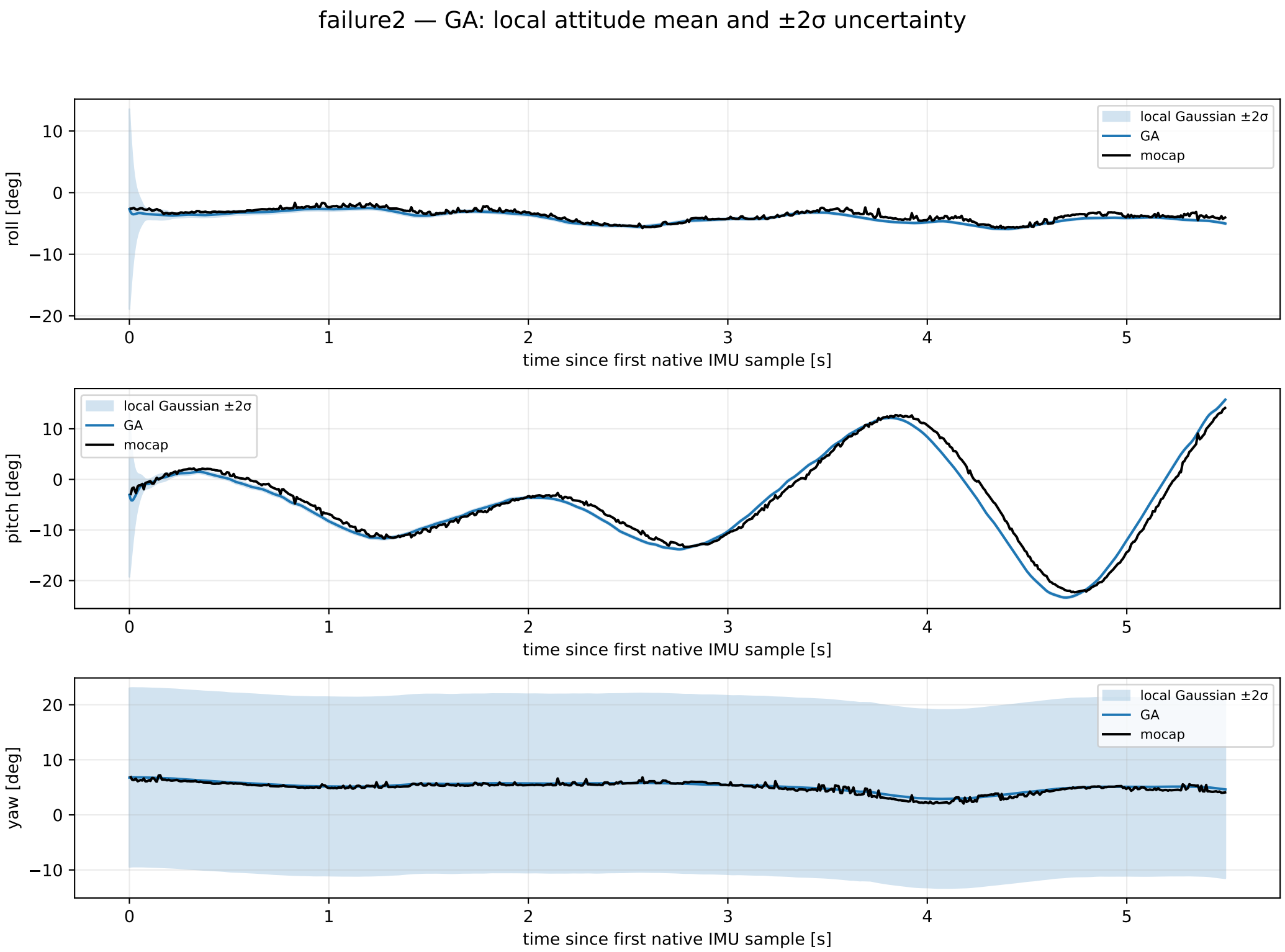

failure2 — Applied residual wrench in reconstructed dynamics (body frame, about CoG)
applied law ±2σ
added residual-wrench mean
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time since first native IMU sample [s]
The native-IMU residual wrench is added to the dynamics; this is not a model-only free replay.

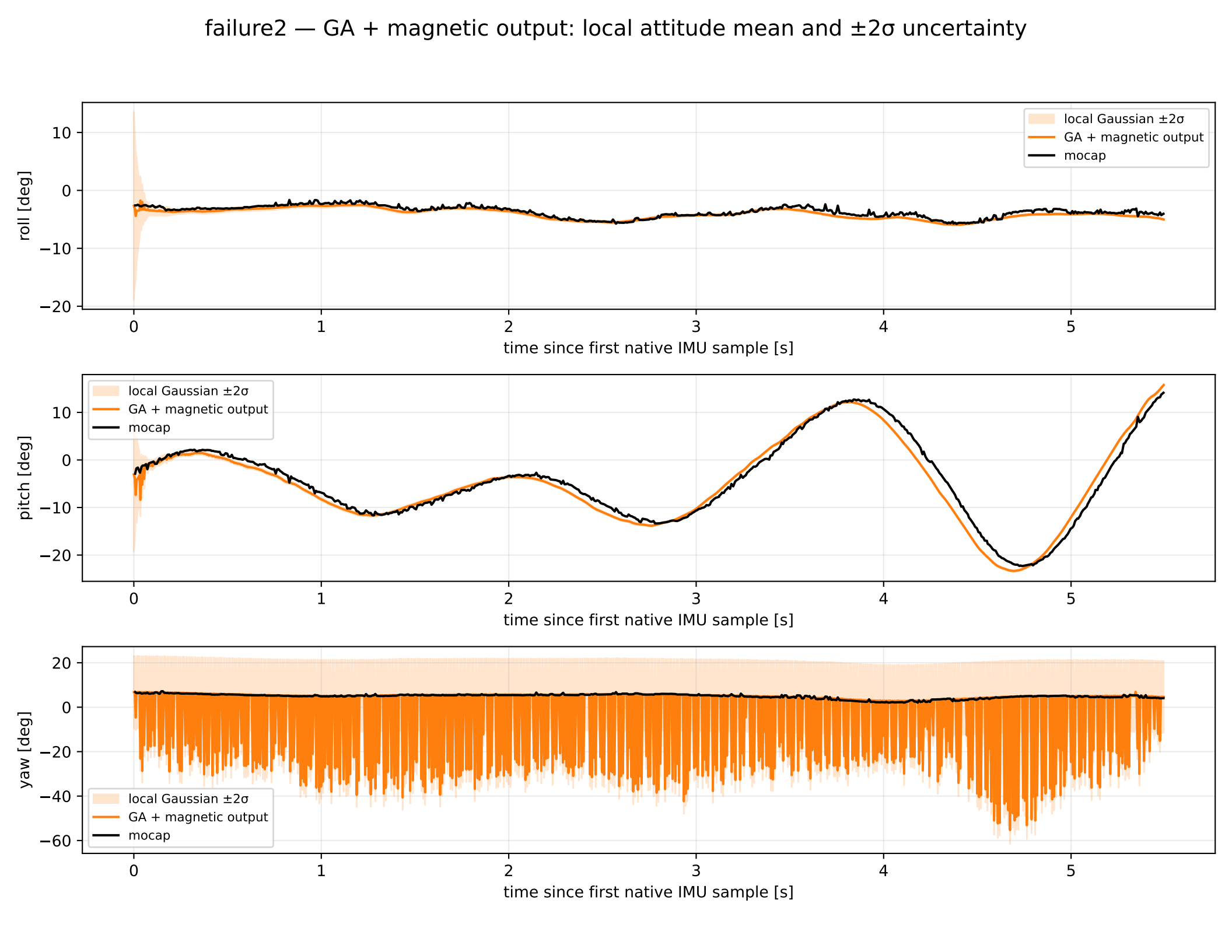

failure2 — gyro: raw, inverse-lag correction and model measurement response
x [rad/s]
y [rad/s]
z [rad/s]
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response
time since first native IMU sample [s]

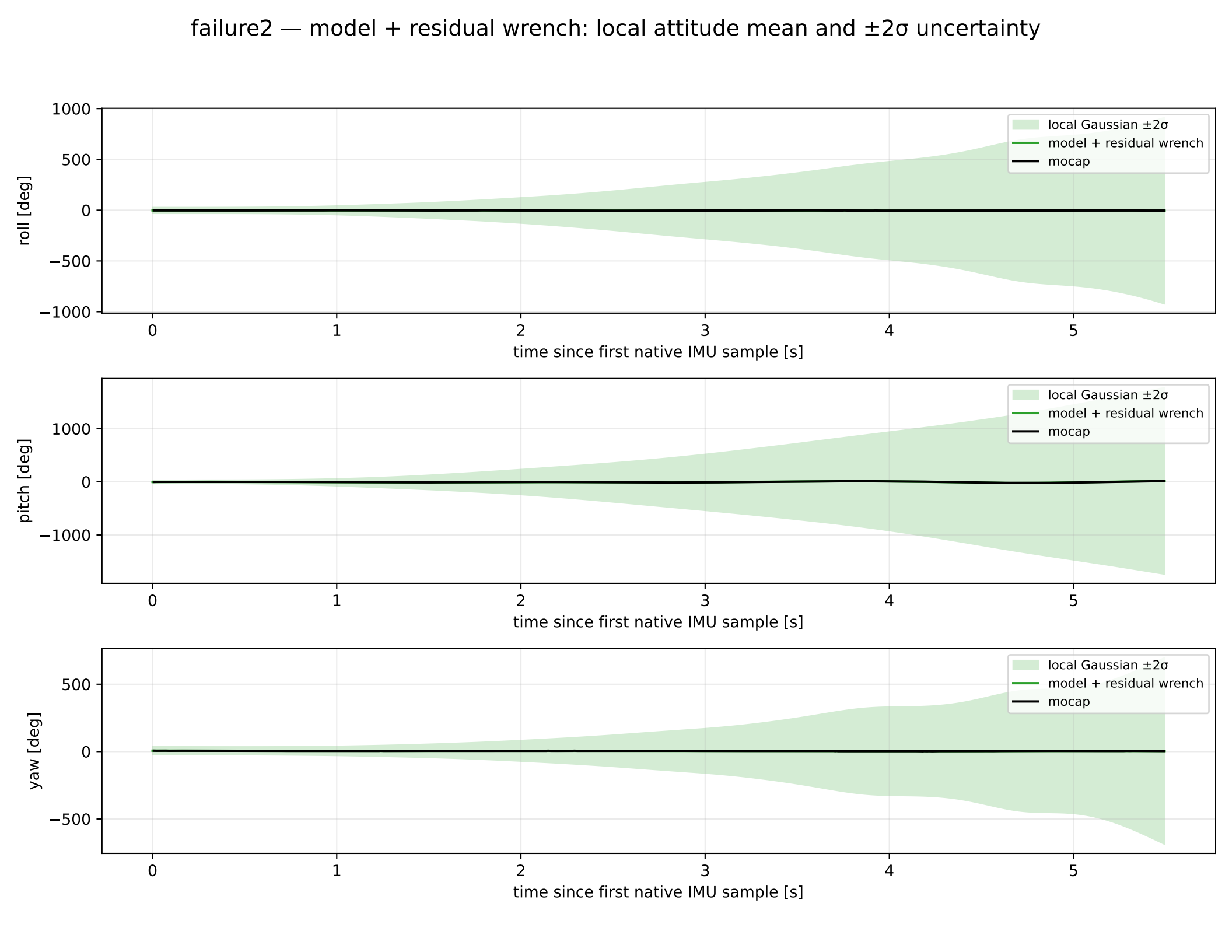

failure2 — accel: raw, inverse-lag correction and model measurement response
x [m/s²]
y [m/s²]
z [m/s²]
time since first native IMU sample [s]
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response


failure2 — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

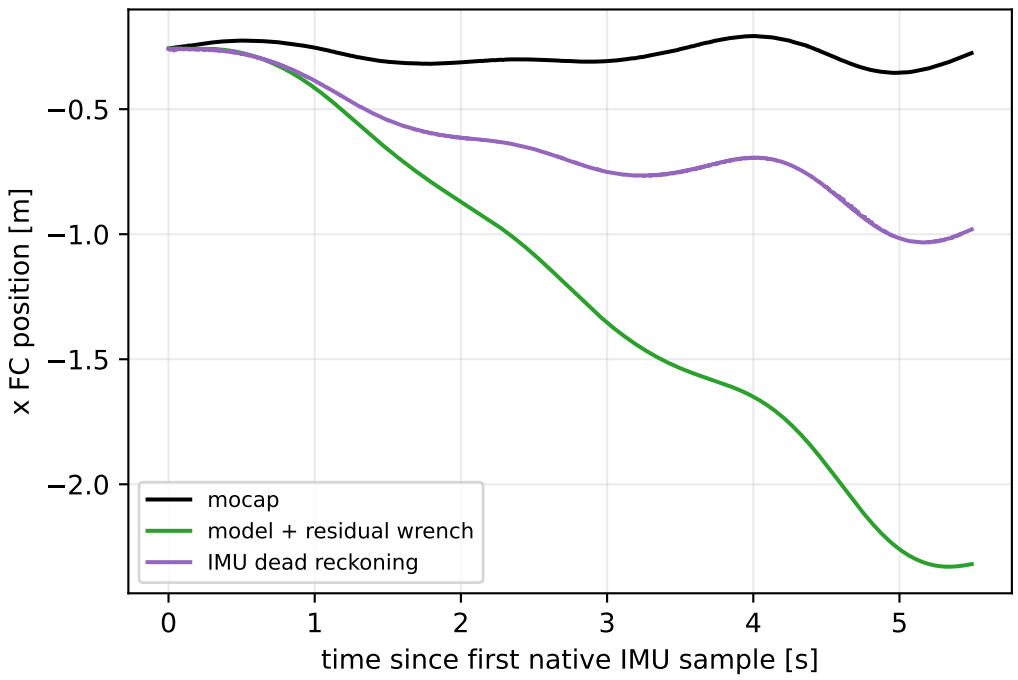

failure2 — SO(3) attitude errors at original mocap timestamps
GA: RMS 1.774°, final 1.997°
GA + magnetic output: RMS 16.910°, final 1.997°
model + residual wrench: RMS 2.017°, final 3.342°
geodesic attitude error [deg]
time since first native IMU sample [s]
0
10
20
30
40
50
0
1
2
3
4
5

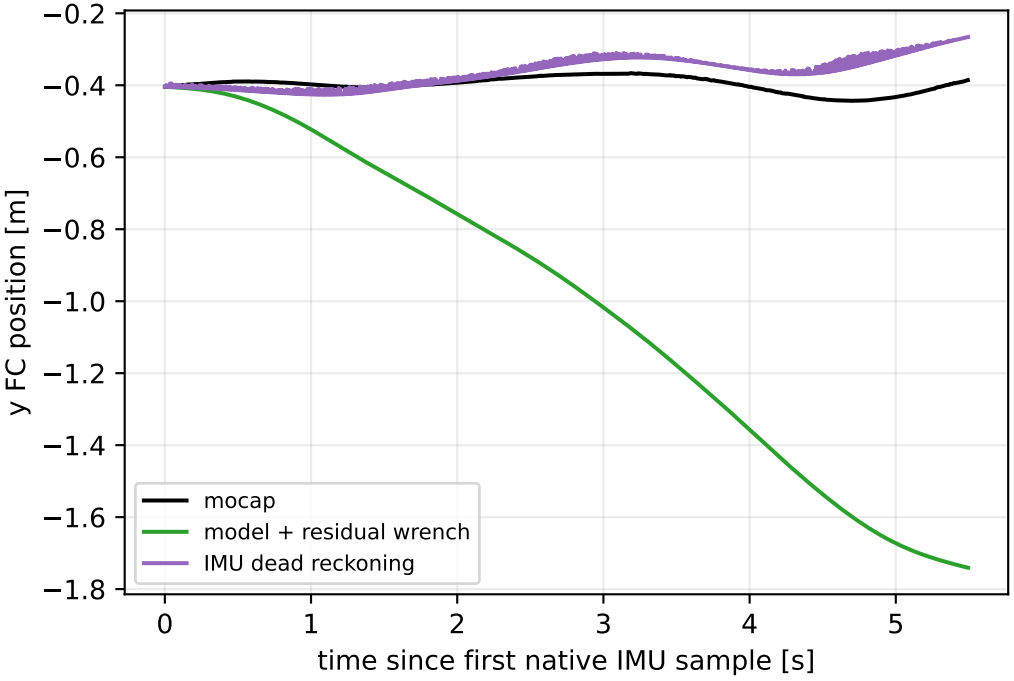

failure2 — GA: local attitude mean and ±2σ uncertainty
local Gaussian ±2σ
GA
mocap
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]

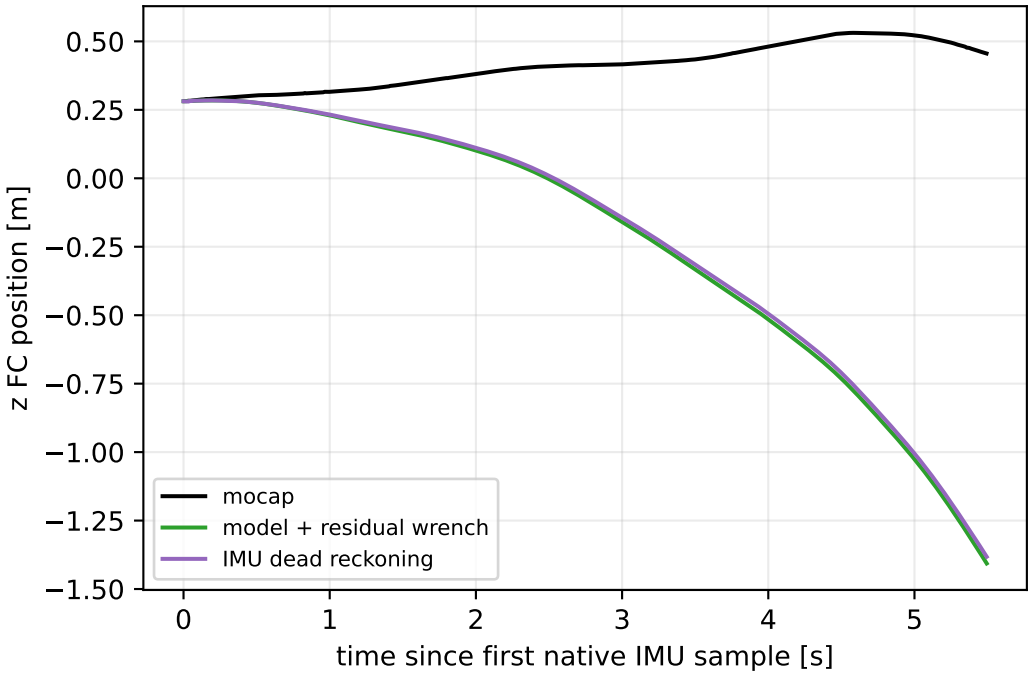

failure2 — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

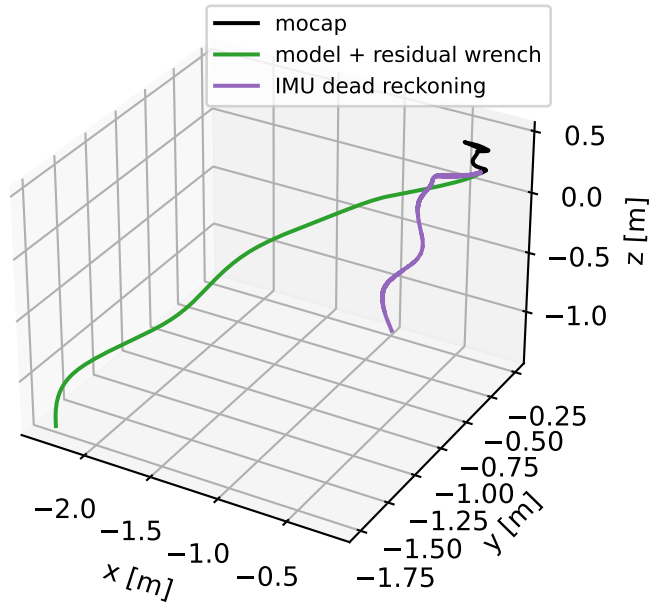

failure2 — model + residual wrench: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
model + residual wrench
mocap

failure2 — Position reconstruction (mean trajectories)



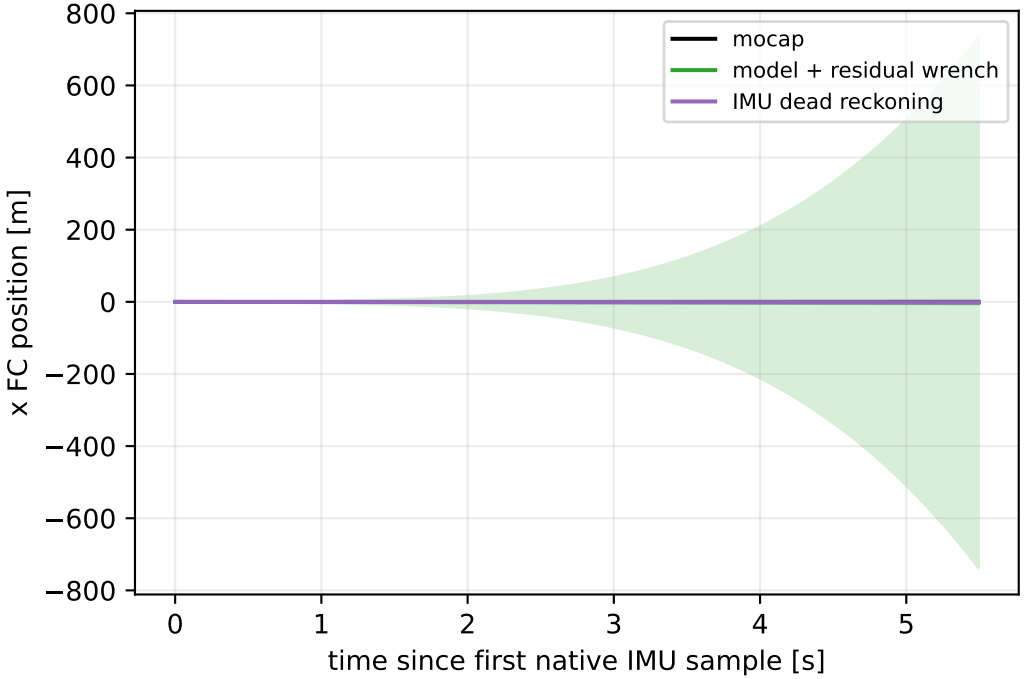


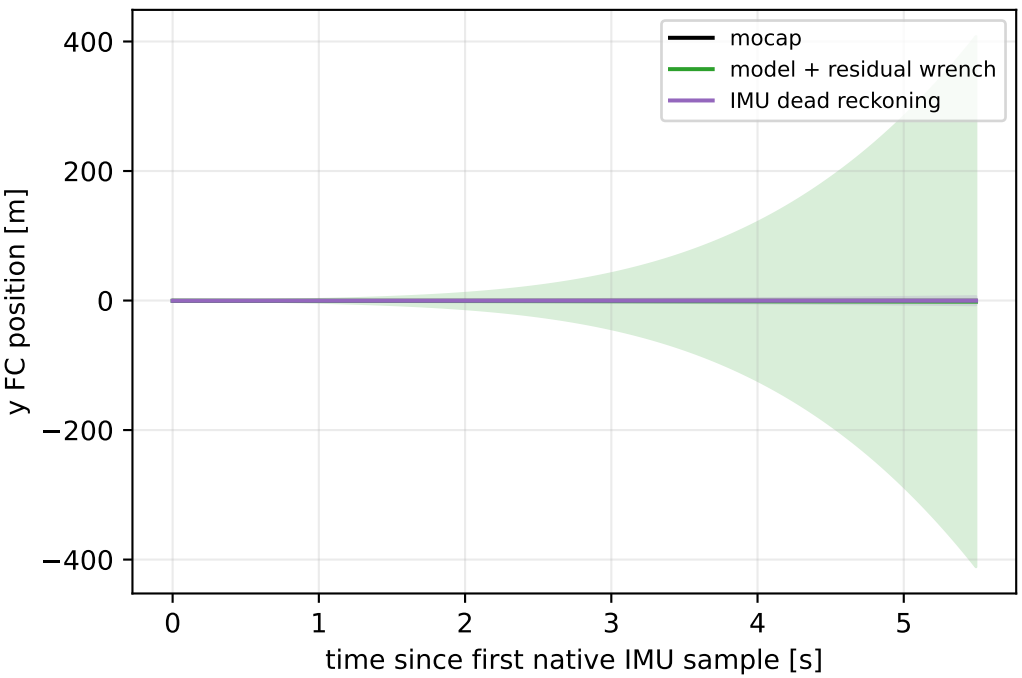


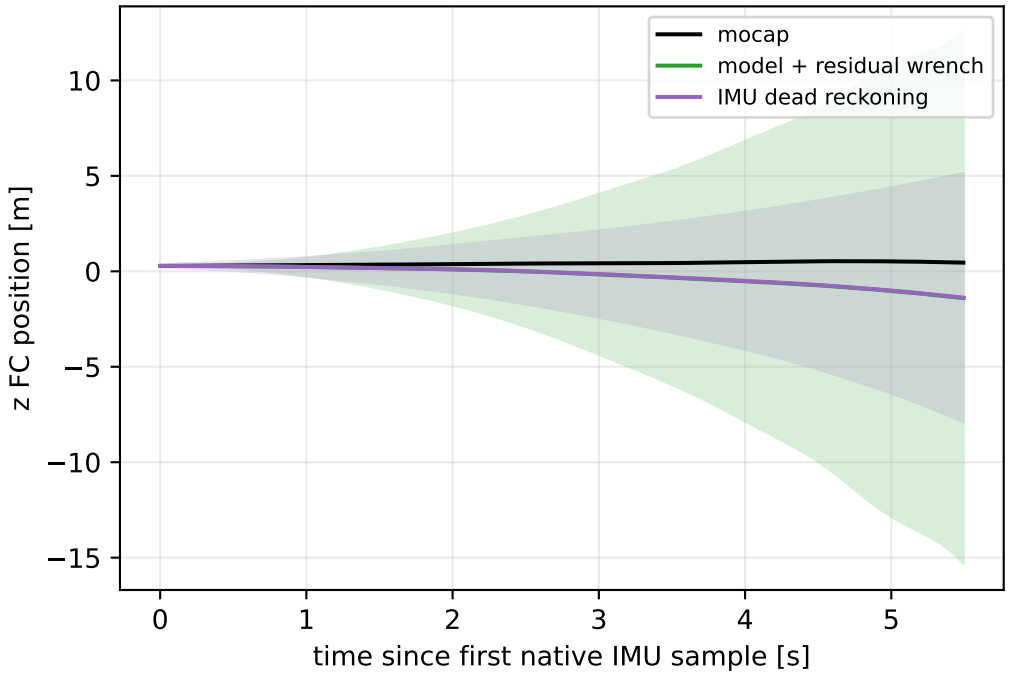


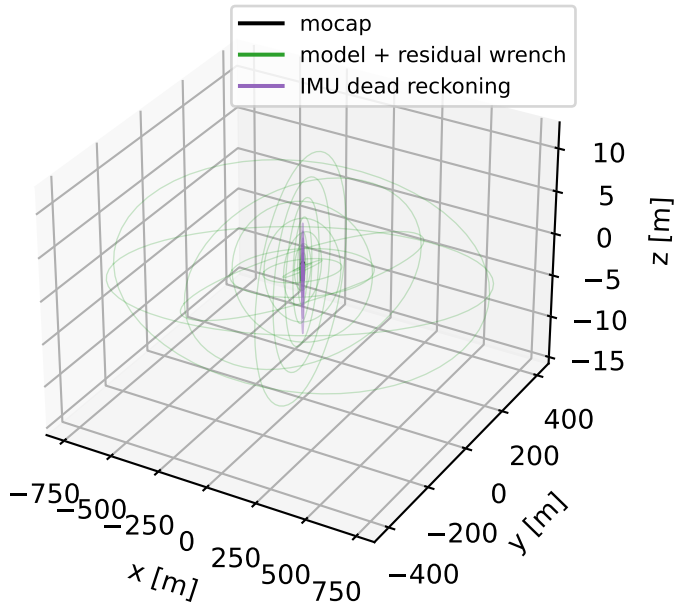
Residual-wrench model replay and IMU integration are not reset to later mocap.

failure2 — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.

failure2 — Actual estimated SI values and joint covariance coordinate order

#	parameter	unit	estimate	std. deviation
1	mass_kg	kg	4.202190638e+00	2.120401021e+03
2	inertia_xx_kg_m2	kg m^2	3.251766764e-01	1.640699217e+02
3	inertia_yy_kg_m2	kg m^2	9.977341905e-02	5.034457425e+01
4	inertia_zz_kg_m2	kg m^2	3.814040225e-01	1.924906717e+02
5	inertia_xy_kg_m2	kg m^2	5.313074270e-03	2.679818991e+00
6	inertia_xz_kg_m2	kg m^2	3.438717982e-02	1.737091680e+01
7	inertia_yz_kg_m2	kg m^2	-4.707781661e-02	2.375683404e+01
8	cog_x_m	m	-4.500179327e-03	2.240183553e-03
9	cog_y_m	m	8.127700298e-03	9.877383018e-04
10	cog_z_m	m	3.331267616e-03	1.050935365e-03
11	force_effectiveness_1	dimensionless	1.521768316e+00	7.678800039e+02
12	force_effectiveness_2	dimensionless	1.412146048e+00	7.125534991e+02
13	force_effectiveness_3	dimensionless	1.607095682e+00	8.109241551e+02
14	force_effectiveness_4	dimensionless	1.487139969e+00	7.504091076e+02
15	thrust_time_constant_seconds	s	3.552099499e-01	6.526905702e-03
16	gyro_time_constant_seconds	s	6.159870505e-02	3.906426236e-03
17	accel_time_constant_seconds	s	6.209106763e-02	8.514447286e-04

Common-scale gauge: physical chart projected onto existing 13-D orthogonal common-scale section; tau coordinates unaffected

Covariance units are products of row/column units. All cross terms and weak modes are retained.

Full-precision values: parameter_values_si.csv; full 17×17 covariance: parameter_covariance_si.csv.

failure2 — Full SI covariance: rows 1–9, columns 1–9

	1	2	3	4	5	6	7	8	9
1	4.496100e+06	3.478940e+05	1.067507e+05	4.081574e+05	5.682290e+03	3.683329e+04	-5.037401e+04	-1.984904e+00	1.288606e-01
2	3.478940e+05	2.691894e+04	8.260030e+03	3.158193e+04	4.396776e+02	2.850044e+03	-3.897782e+03	-1.536145e-01	9.986693e-03
3	1.067507e+05	8.260030e+03	2.534576e+03	9.690861e+03	1.349143e+02	8.745310e+02	-1.196028e+03	-4.713276e-02	3.062253e-03
4	4.081574e+05	3.158193e+04	9.690861e+03	3.705266e+04	5.158401e+02	3.343738e+03	-4.572969e+03	-1.801826e-01	1.169282e-02
5	5.682290e+03	4.396776e+02	1.349143e+02	5.158401e+02	7.181430e+00	4.655089e+01	-6.366400e+01	-2.509028e-03	1.629895e-04
6	3.683329e+04	2.850044e+03	8.745310e+02	3.343738e+03	4.655089e+01	3.017488e+02	-4.126778e+02	-1.628059e-02	1.066597e-03
7	-5.037401e+04	-3.897782e+03	-1.196028e+03	-4.572969e+03	-6.366400e+01	-4.126778e+02	5.643872e+02	2.224296e-02	-1.446684e-03
8	-1.984904e+00	-1.536145e-01	-4.713276e-02	-1.801826e-01	-2.509028e-03	-1.628059e-02	2.224296e-02	5.018422e-06	-1.564380e-06
9	1.288606e-01	9.986693e-03	3.062253e-03	1.169282e-02	1.629895e-04	1.066597e-03	-1.446684e-03	-1.564380e-06	9.756270e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure2 — Full SI covariance: rows 1–9, columns 10–17

	10	11	12	13	14	15	16	17
1	2.220203e-01	1.628214e+06	1.510899e+06	1.719484e+06	1.591168e+06	9.598091e-01	1.770249e+00	-7.991522e-01
2	1.716198e-02	1.259860e+05	1.169086e+05	1.330483e+05	1.231196e+05	7.424657e-02	1.369361e-01	-6.183330e-02
3	5.266943e-03	3.865859e+04	3.587320e+04	4.082563e+04	3.777903e+04	2.277687e-02	4.202278e-02	-1.897410e-02
4	2.015259e-02	1.478097e+05	1.371599e+05	1.560953e+05	1.444468e+05	8.715126e-02	1.606854e-01	-7.254145e-02
5	2.803725e-04	2.057779e+03	1.909514e+03	2.173130e+03	2.010960e+03	1.210109e-03	2.238112e-03	-1.010058e-03
6	1.809200e-03	1.333877e+04	1.237770e+04	1.408649e+04	1.303529e+04	7.835290e-03	1.448805e-02	-6.547959e-03
7	-2.485659e-03	-1.824240e+04	-1.692801e+04	-1.926499e+04	-1.782734e+04	-1.075491e-02	-1.982759e-02	8.954450e-03
8	1.082730e-06	-7.188254e-01	-6.669849e-01	-7.590842e-01	-7.024953e-01	2.916406e-06	1.236765e-06	8.418126e-07
9	-6.008383e-07	4.666920e-02	4.328644e-02	4.927784e-02	4.561924e-02	-8.947103e-07	-1.231535e-06	-2.808812e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure2 — Full SI covariance: rows 10–17, columns 1–9

	1	2	3	4	5	6	7	8	9
10	2.220203e-01	1.716198e-02	5.266943e-03	2.015259e-02	2.803725e-04	1.809200e-03	-2.485659e-03	1.082730e-06	-6.008383e-07
11	1.628214e+06	1.259860e+05	3.865859e+04	1.478097e+05	2.057779e+03	1.333877e+04	-1.824240e+04	-7.188254e-01	4.666920e-02
12	1.510899e+06	1.169086e+05	3.587320e+04	1.371599e+05	1.909514e+03	1.237770e+04	-1.692801e+04	-6.669849e-01	4.328644e-02
13	1.719484e+06	1.330483e+05	4.082563e+04	1.560953e+05	2.173130e+03	1.408649e+04	-1.926499e+04	-7.590842e-01	4.927784e-02
14	1.591168e+06	1.231196e+05	3.777903e+04	1.444468e+05	2.010960e+03	1.303529e+04	-1.782734e+04	-7.024953e-01	4.561924e-02
15	9.598091e-01	7.424657e-02	2.277687e-02	8.715126e-02	1.210109e-03	7.835290e-03	-1.075491e-02	2.916406e-06	-8.947103e-07
16	1.770249e+00	1.369361e-01	4.202278e-02	1.606854e-01	2.238112e-03	1.448805e-02	-1.982759e-02	1.236765e-06	-1.231535e-06
17	-7.991522e-01	-6.183330e-02	-1.897410e-02	-7.254145e-02	-1.010058e-03	-6.547959e-03	8.954450e-03	8.418126e-07	-2.808812e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure2 — Full SI covariance: rows 10–17, columns 10–17

	10	11	12	13	14	15	16	17
10	1.104465e-06	8.039873e-02	7.462070e-02	8.491301e-02	7.856211e-02	1.369631e-06	2.631578e-06	-6.290448e-08
11	8.039873e-02	5.896397e+05	5.471556e+05	6.226924e+05	5.762241e+05	3.475666e-01	6.410718e-01	-2.894063e-01
12	7.462070e-02	5.471556e+05	5.077325e+05	5.778268e+05	5.347066e+05	3.225737e-01	5.949057e-01	-2.685470e-01
13	8.491301e-02	6.226924e+05	5.778268e+05	6.575980e+05	6.085249e+05	3.670796e-01	6.770175e-01	-3.056261e-01
14	7.856211e-02	5.762241e+05	5.347066e+05	6.085249e+05	5.631138e+05	3.396533e-01	6.264697e-01	-2.828245e-01
15	1.369631e-06	3.475666e-01	3.225737e-01	3.670796e-01	3.396533e-01	4.260050e-05	-1.272511e-05	-2.385254e-07
16	2.631578e-06	6.410718e-01	5.949057e-01	6.770175e-01	6.264697e-01	-1.272511e-05	1.526017e-05	3.588101e-08
17	-6.290448e-08	-2.894063e-01	-2.685470e-01	-3.056261e-01	-2.828245e-01	-2.385254e-07	3.588101e-08	7.249581e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.